International Journal of Pattern Recognition
and Artificial Intelligence

Feedback-Controlled Diversity Regularization for Multi-Head Attention: Enforcing a Similarity Budget Late in Training

Kejie Hu
City University of Wu Han, Wuhan Hubei 430083, CN
xshu1999@gmail.com

Received (10 September 2026)

Multi-head attention tends to learn redundant heads; training-time diversity penalties reduce them, but their weight is hard to set. A systematic Cora sweep shows no usable operating point: weights ≤ 0.5 leave inter-head similarity essentially unchanged (0.95–0.98), while weights ≥ 5 move it substantially (0.37–0.67) at a cost of 0.7–2.9 accuracy points, the weakest weight ($\lambda = 5$) doing so at the best checkpoint. We introduce a feedback controller treating a similarity budget as a set point, adjusting the regularization weight per epoch via integral control. On three citation networks with 2-layer GATs (10 seeds), accuracy stays within ± 0.4 points of baseline while the controller meets the budget late in training on Cora (0.71), approaches it on Citeseer (0.85), and barely on Pubmed. It beats GradNorm on two of three datasets; linear and Kendall schedules match it. As a pruning criterion it matches standard criteria, a null surviving a revival test at late checkpoints, and KD after pruning recovers 82.00 ± 0.69 on Cora, not significantly different from students trained from scratch with KD at equal budgets, fine-tuning alone leaving a 2.1–2.4 point gap. We claim no accuracy gains from the regularizer or criterion superiority, on the datasets we test.

Keywords: multi-head attention; diversity regularization; feedback control; head pruning; knowledge distillation; graph attention networks

1. Introduction

Multi-head attention has measurable redundancy. A large fraction of attention heads can be removed after training with little change in accuracy,[14] and similarity between attention patterns is a common proxy for that redundancy.[1] Two lines of work exploit this observation. One line prunes or merges heads *after* training using importance or similarity proxies.[1,6,14,18] Another line adds a *training-time* diversity penalty so that heads become less similar in the first place.[2,15] Both lines use fixed hyperparameters: a fixed pruning ratio or a fixed regularization weight.

In this paper we study the regularization weight itself. Our starting point is an empirical observation about fixed weights, obtained from a systematic sweep on Cora (Table 1). A diversity penalty that penalizes the cosine similarity between attention matrices of different heads has two failure modes, and no intermediate weight escapes them. Weights at or below 0.5 leave inter-head similarity essentially untouched (0.95–0.98) and $\lambda = 1$ moves it only slightly (0.91); weights at or above 5 do reduce similarity (to 0.37–0.67) but cost about 0.7–2.9 points of test accuracy. The weakest effective weight ($\lambda = 5$) reduces similarity already at the best checkpoint (0.67), i.e. during the accuracy-critical phase. There is no fixed weight that is both effective and harmless.

This failure spectrum motivates treating the regularization weight as a *controlled variable* rather than a tuned hyperparameter. We propose a feedback controller: each epoch, the current inter-head similarity is compared against a target budget, and the regularization weight is adjusted by an integral control law. The controller enforces the budget late in training, the phase where fixed weights are either too weak or too strong, while test accuracy remains within ± 0.4 points of the unregularized baseline on all three datasets. The budget is met on Cora (late similarity 0.71), partially met on Citeseer (0.85), and barely moved on Pubmed within the 200-epoch budget (Sec. 4.2); we discuss this dataset dependence explicitly in Sec. 5.

Because the regularization signal is itself a redundancy measure, the same metric can serve as a pruning criterion: after training with the adaptive regularizer, we score heads by their pairwise similarity to other heads and keep the least-redundant subset. We then study how this criterion interacts with fine-tuning and knowledge distillation (KD). We report two findings. First, KD restores most of the pruning loss (+2.1–2.4 points over fine-tuning alone, $p = 0.001$) and shows no significant difference from small students trained from scratch with KD at identical budgets: the structured student reaches the accuracy of the from-scratch student through the pruning path, with no wall-time saving (Cora; the other two datasets in Appendix C). Second, a strict paired comparison of pruning criteria (similarity vs. gradient, random, and magnitude) finds no reliable difference among them at GAT scale: 100 epochs of fine-tuning erase criterion choice. We report this as a negative result, consistent with the caution in[14] that criterion quality must be judged after retraining, not before.

Contributions.

1. A feedback-controlled weighting scheme for head-diversity regularization that enforces a similarity budget late in training without accuracy loss; the budget is met on Cora, partially met on Citeseer, and response-limited on Pubmed; in a regime where no fixed weight is both effective and harmless, and against adaptive baselines (linear, Kendall-style, and GradNorm schedules; 3 datasets $\times$ 10 seeds; full statistics in Sec. 4.2, Table 2).
2. A structured-compression protocol that reuses the diversity metric as the pruning criterion: with KD, pruned students show no significant difference from pure-KD students at matched head budgets, and KD recovers +2.1–2.4 points over fine-tuning alone ($p = 0.001$, Cohen's $d \approx 2.1$; Cora in the main text, Citeseer/Pubmed extension in Appendix C).
3. A strict negative result on pruning-criterion choice at GAT scale (paired Wilcoxon signed-rank tests over identical seeds, effect sizes, multiplicity dis-

cussion), plus a mechanism consistent with the data: at the best checkpoint, head similarity is high for every configuration, so the similarity signal does not differentiate heads at the point where pruning happens. A revival test with criteria computed from late-training checkpoints, where the signal is differentiated, reproduces the null (Table 4).

4. Reproducible Colab-scale artifacts (T4, <100k parameters; protocol, seeds, and code at `https://github.com/HuKejie/div-prune`).

Explicit non-claims. We do not claim that the regularizer *improves* accuracy (it is no-harm only); we do not claim that the diversity criterion is better than existing pruning criteria (the contrast is null); we do not claim that pruning+KD *beats* pure KD (no significant difference is detectable, Sec. 4.4). All evidence is at GAT scale on three citation networks; transfer to other architectures is untested.

2. Related Work

Head redundancy and pruning. The redundancy of multi-head attention is well documented.[14] shows that many heads can be removed with little loss and proposes a greedy, gradient-based head-importance criterion, cautioning that pruning without retraining misrepresents criterion quality.[18] learns L0 gates during training to prune heads. At LLM scale, similarity between attention patterns is exploited for inference-time head clustering and fusion: CHAI clusters correlated heads,[1] DHA fuses similar key/value heads,[6] and ProxyAttn approximates attention with representative heads.[20] These works use head similarity *after* training; none uses it as a training-time regularization signal. Note the scale gap: that evidence is at LLM scale, while all experiments in this paper are at 92k-parameter GAT scale on three citation networks (Sec. 5).

Diversity regularization for attention. Training-time diversity penalties exist in several forms: HSIC or orthogonality regularizers on head subspaces,[23] diversity-promoting losses for ASR conformers,[3] multi-level redundancy reduction for ViTs,[5] and orthogonalization of head input gradients.[15] DEACON trains a GHA-based projection so that heads occupy a maximum-variance subspace and then prunes heads with the same PCA criterion,[2] the closest work coupling diversity-enforcing training with same-criterion pruning. It differs from ours in criterion (PCA variance vs. pairwise cosine similarity), in weighting (fixed vs. feedback-controlled), and in scope (no GNN coverage, no distillation).[11] adapts the strength of an entropy-based diversity regularizer per head for private inference; to our knowledge no prior work applies feedback control to the global weight of a diversity loss.

Adaptive loss weighting. Learning or scheduling loss weights is standard in multi-task learning: homoscedastic uncertainty weighting,[13] gradient normalization,[7] and excess-risk weighting.[98] learns the strength of a regularization term in a variational fine-tuning setting. These mechanisms target task or regularization balance; none is applied to head-diversity regularization.

KD and GNN compression. KD[10] compresses models by matching student outputs to teacher outputs. For transformers, TinyBERT and MiniLM distill attention-level information;[12,19] for GNNs, GLNN distills a GNN teacher into an MLP student[24] and KRD shows reliability-weighted distillation can exceed the teacher.[21] Pruning-before-distillation pipelines exist,[4,16,25] but none couples a diversity criterion with distillation. Our closest comparison is pure KD at matched budgets (Sec. 4.4); omitting it would leave open the possibility that KD alone already captures the benefit.

Positioning. The combination we study, (a) the same similarity metric used for training-time regularization and for pruning, (b) a feedback-controlled regularization weight, and (c) distillation, at small-model scale (GAT, ≤100k parameters), is, to our knowledge, unoccupied. The nearest neighbors are DEACON[2] (shared criterion, fixed weights, no KD) and Nicolicioiu et al.[15] (regularize then prune, but the pruning signal is OOD attribution, not the diversity metric).

3. Method

3.1. *Setup and diversity regularization*

We use a standard 2-layer GAT[17] with 8 heads and hidden size 8 per head (92,373 parameters) on the Planetoid citation networks Cora, Citeseer, Pubmed.[22] Let $A_h^{(l)}$ denote the attention matrix of head h in layer l (softmax-normalized attention coefficients). The diversity loss penalizes the mean pairwise cosine similarity between attention matrices of different heads in the same layer:

$$\mathcal{L}_{\text{div}} \;=\; \frac{2}{H(H-1)} \sum_{l} \sum_{h<k} \cos\big(A_h^{(l)},\, A_k^{(l)}\big). \tag{1}$$

The training objective is $L = L_{\text{CE}} + \lambda \mathcal{L}_{\text{div}}$, where L_{CE} is the cross-entropy loss. This is the attention-level variant; an output-level variant (Euclidean distance between head outputs) exists but is not used in the main experiments.

3.2. *The fixed-λ failure spectrum*

Sweeping λ on Cora (Table 1) shows two failure modes. For $\lambda \leq 0.5$, the penalty is too weak to move similarity: inter-head similarity stays at 0.95–0.98, indistinguishable from unregularized training, and $\lambda = 1$ moves it only slightly (0.91). For $\lambda \geq 5$, similarity drops (0.37–0.67 at the best checkpoint) but test accuracy degrades by about 0.7–2.9 points. The weakest effective weight, $\lambda = 5$, already reaches 0.67 *at the best checkpoint*: it enforces diversity during the accuracy-critical phase, which is the timing failure we attribute to GradNorm in Sec. 4.2. No fixed weight is simultaneously effective and harmless.

3.3. *Feedback-controlled adaptive λ*

We treat the target similarity budget d^* as a set point and λ as the control variable. Each epoch t, we compute the current similarity d_t (the same quantity as in Eq. (1), averaged over the training set) and update

$$\lambda_{t+1} \;=\; \mathrm{clamp}_{[0,\,\lambda_{\max}]}\big(\lambda_t + \eta\,(d_t - d^*)\big), \tag{2}$$

starting from $\lambda_0 = 0$, with target $d^* = 0.7$, gain $\eta = 0.05$, and ceiling $\lambda_{\max} = 3.0$, held constant across all three datasets. The integral law accumulates error only while d_t exceeds the budget, so λ winds up when similarity is high and stops changing once the budget is met. No λ is set by hand: the same three constants $(d^*, \eta, \lambda_{\max})$ are used everywhere. The three constants were chosen once on Cora and never re-tuned: $d^* = 0.7$ sits just above the similarities the fixed sweep reaches (0.37–0.67) and far below the unregularized level ($\sim$0.97); $\eta = 0.05$ gives a per-epoch weight increment of $\eta\,(d_t - d^*) \approx 0.015$ at the unregularized similarity, so the weight builds up over the $\sim$100-epoch late phase rather than overshooting within a few epochs; and $\lambda_{\max} = 3.0$ sits below the regime ($\lambda \geq 5$) where fixed weights become harmful (Sec. 4.1). A sensitivity grid over (η, d^*) is reported in Appendix B. The ceiling is a stability guard, not an operating point: λ converges below it on every run (final values 0.27–1.94 per dataset at the chosen gain; a 4×-gain probe on Pubmed saturates the ceiling, Sec. 4.2). The update is cheap (one forward-pass statistic per epoch) and adds no parameters.

3.4. *Pruning with the diversity criterion*

After training with the adaptive regularizer, the diversity metric doubles as a pruning criterion. For each head, we compute its mean pairwise cosine similarity to all other heads in the same layer (over the training set); heads with the lowest similarity are the least redundant and are kept. The pruning protocol follows the structure of:[14] equal fractions per layer, at least one head per layer kept, with prune ratios 0.5 and 0.75 (keeping 4/8 and 2/8 heads). Baselines are gradient-based importance,[14] random selection, and magnitude (L_2 norm of head weights). The criterion is scored at a checkpoint of the trained teacher: the early-stopping best checkpoint, the lowest-similarity checkpoint, or the final checkpoint; the main results use the best checkpoint (the revival test across sources is reported in Sec. 4.3). Pruned models are then fine-tuned for a fixed 100 epochs.

3.5. *KD pipeline*

For the compression study, the pruned student is trained by logit KD from the dense adaptive teacher: $L = (1-\alpha)\,L_{\mathrm{CE}} + \alpha\,T^2\,L_{\mathrm{KL}}$, with $\alpha = 0.5$ and $T = 2$, for 100 epochs, the same epoch budget as the pure-KD comparison (students trained from scratch with KD). All comparisons are at matched head budgets (4/8 or 2/8 heads) so that the accuracy–parameter trade-off, not the training recipe, is the variable. Because both GAT layers shrink with the head count (layer 1 per-head attention weights, layer 2 concatenation input), matched head budgets also match parameter counts: 92,373 parameters at 8/8 heads, 46,197 at 4/8, and 23,109 at 2/8. In addition to the three arms B5/B6/B8 (pure KD, prune + fine-tuning, prune + KD), we report B4: small models trained from scratch at the same budgets with plain cross-entropy and no distillation, the reference that tells us whether the pruning path loses anything relative to simply training small.

4. Experiments

Setup and statistics. GAT 2-layer, 8 heads, hidden 8/head (92,373 parameters), Adam optimizer (learning rate 0.005, weight decay $5{\times}10^{-4}$), dropout 0.6, fixed Planetoid split, early stopping with patience 100 (max 200 epochs), fine-tuning and KD for 100 epochs. 10 seeds per configuration. Results are mean $\pm$ std over seeds; 95% CIs are in the appendix (Table 6 and Table 7). Comparisons use paired Wilcoxon signed-rank tests over identical seeds: one-sided where a direction is hypothesized, two-sided with 95% CIs on the difference for tie claims; where tied pairs occur, the normal approximation is used and the paired t-test is reported alongside. Effect sizes (Cohen's d) are reported for major contrasts. Compute: Google Colab T4; each dense run takes seconds at this scale (wall times in Sec. 4.4). The revision experiments (adaptive baselines, controller checkpoints, the revival matrix, and the H3 extension) were executed from a versioned Colab notebook with per-combination progress logging; merged results and teacher checkpoints are archived (Sec. 5). The second-revision experiments ($\lambda = 5$ comparisons and the controller sensitivity grid) were executed locally with the same Hydra pipeline and identical configs; their CSV results are archived with the same layout.

4.1. *Can fixed λ control diversity?*

Claim tested: no fixed weight is simultaneously effective and harmless.

Table 1 (top) reports the Cora sweep. The answer is no: $\lambda \leq 0.5$ leaves similarity at 0.95–0.98 and $\lambda = 1$ moves it only slightly (0.91); $\lambda \geq 5$ moves it but costs about 0.7–2.9 points. Note where $\lambda = 5$ moves it: at the best checkpoint (0.67), not late in training, the timing failure analyzed in Sec. 4.2. This is the motivation for feedback control.

4.2. *Adaptive λ vs. fixed λ vs. baseline*

Claim tested: the controller enforces the budget without accuracy loss where fixed λ fails.

Table 1 (bottom) and Fig. 1 compare baseline, fixed $\lambda = 20$, and adaptive ($t = 0.7$, $\eta = 0.05$) on all three datasets (fixed $\lambda = 5$ is examined through the sweep above). Adaptive λ keeps accuracy within ± 0.4 points of baseline everywhere (Cora 80.79 ± 0.92 vs. 81.15 ± 1.02; Citeseer 68.46 ± 1.35 vs. 68.27 ± 1.56; Pubmed 77.44 ± 0.67 vs. 77.52 ± 0.54); all CIs overlap. Fixed $\lambda = 20$ is below both everywhere (-1.1 to -1.9). Fixed $\lambda = 5$, the strongest fixed weight in the sweep, costs about 0.7 points on Cora and moves similarity at the best checkpoint (details in Table 1): it belongs to the same timing quadrant as GradNorm (below). The controller converges to

Table 1. Fixed-λ failure spectrum on Cora (top) and the main comparison of training configurations across three citation networks (bottom). Accuracy is mean $\pm$ std over seeds. Top: second-revision re-run at n=10 per weight with recorded similarities, which supersedes the pre-submission n=5 sweep. Bottom: n=10; the Cora fixed-λ=20 cell uses the second-revision batch, Citeseer and Pubmed use the original batch (identical protocol). d_{best} is the inter-head similarity at the best checkpoint.

Fixed-λ sweep, Cora (attention-level)			
λ	Test acc. (%)	d_{best}	
0.01	81.16 ± 1.63	0.979	
0.1	81.03 ± 1.50	0.978	
0.5	81.13 ± 1.37	0.954	
1.0	81.08 ± 1.68	0.906	
5.0	80.50 ± 1.44	0.674	
20.0	79.71 ± 1.24	0.597	
50.0	78.22 ± 1.88	0.366	
Main comparison (10 seeds per cell)			
Dataset	Baseline	Fixed λ=20	Adaptive (t=0.7, η=0.05)
Cora	81.15 ± 1.02	79.71 ± 1.24	80.79 ± 0.92
Citeseer	68.27 ± 1.56	66.55 ± 2.00	68.46 ± 1.35
Pubmed	77.52 ± 0.54	76.30 ± 0.68	77.44 ± 0.67

different final weights per dataset (Cora $\lambda_{\text{final}} = 0.84$, Citeseer 0.27, Pubmed 1.94), which is the point: the controller finds the per-dataset operating point that a fixed sweep cannot.

Mechanism (Fig. 2). Training traces show the closed loop working: similarity falls from $\sim$0.995 toward the 0.7 budget as the integral controller raises λ, while validation accuracy stays flat. The divergence is enforced *late* in training, after the best checkpoint is selected: the best-checkpoint similarity therefore remains high (Cora 0.934; Citeseer 0.988; Pubmed 0.996). We emphasize this distinction throughout: the controller enforces the budget late in training; it does not claim to reduce similarity at the selected checkpoint. How far the late phase gets is dataset-dependent: on Cora the budget is met (late similarity 0.706); on Citeseer similarity moves substantially toward it (0.847); on Pubmed it barely moves (0.988). A second-revision probe quadrupled the gain ($\eta = 0.2$, nine seeds): the weight saturates at the 3.0 ceiling, yet late similarity still barely moves (0.993). The Pubmed run is thus *response-limited*: its similarity responds too slowly to the penalty at any weight within the configured ceiling, so more gain does not buy the budget there (Appendix B).

Adaptive baselines (Table 2, Fig. 3). A controller that beats only fixed weights invites the question whether any adaptive scheme would do. We compare the controller against three adaptive baselines on all datasets (10 paired seeds): an open-loop linear ramp calibrated to the controller's slope on Cora and transferred unchanged, Kendall–Gal uncertainty weighting,[13] and GradNorm.[7] The

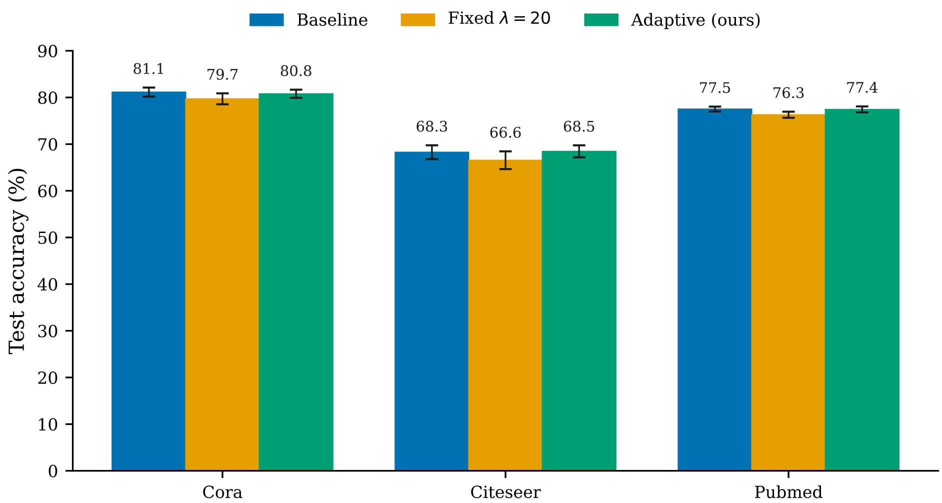

Fig. 1. Test accuracy of a 2-layer GAT on three citation networks under baseline training, fixed-λ ($\lambda = 20$) diversity regularization, and feedback-controlled adaptive regularization (target 0.7, $\eta = 0.05$). Bars are means over 10 seeds; error bars are ± 1 standard deviation over seeds; values above bars are means. The adaptive configuration stays within ± 0.4 points of baseline on all three datasets; no fixed weight achieves both an effect on similarity and this level of accuracy (see Table 1).

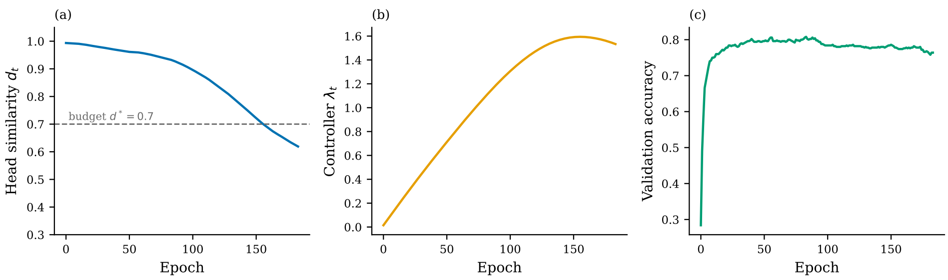

Fig. 2. Adaptive-controller dynamics on Cora, seed 3 (same-protocol run; see Sec. 5 for the provenance note): (a) head similarity d_t falls toward the 0.7 budget as (b) the integral controller raises λ; (c) validation accuracy remains flat. The budget is enforced late in training, after the best checkpoint is selected.

controller matches the linear and Kendall-style schedules on accuracy everywhere ($|\Delta| \leq 0.48$ pp; no contrast is significant under one-sided paired tests) and beats GradNorm significantly on Cora (+1.75 pp, $p = 0.010$) and Pubmed (+1.37 pp, $p = 0.005$; Citeseer +1.10 pp, $p = 0.053$). The difference is timing. GradNorm is the only baseline whose best-checkpoint similarity reaches the 0.7 budget ($d_{\text{best}} = 0.69$ on Cora and Citeseer): it diverges heads during the accuracy-critical phase while

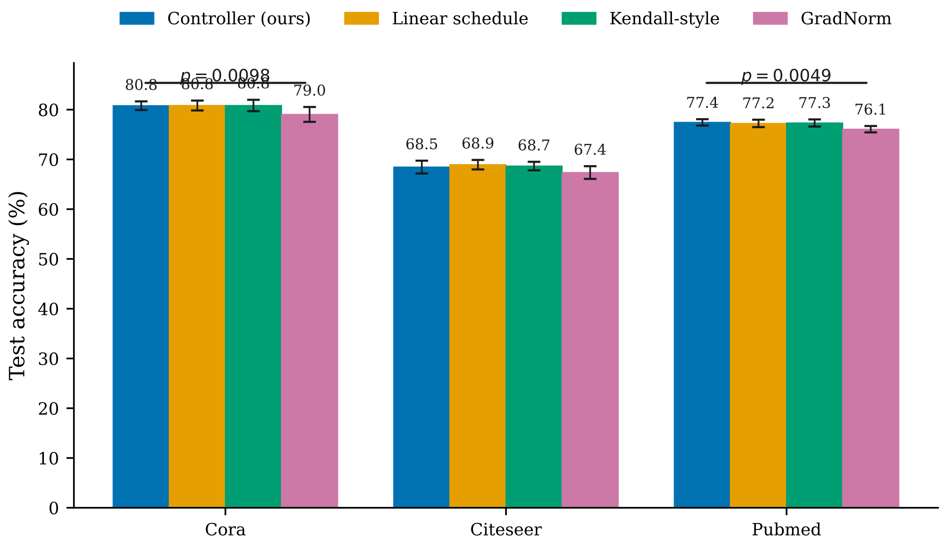

Fig. 3. Controller vs. adaptive baselines on three citation networks. Bars are means over 10 seeds; error bars are ± 1 standard deviation. Brackets: one-sided paired Wilcoxon signed-rank tests, controller vs. GradNorm, $p = 0.010$ (Cora) and $p = 0.005$ (Pubmed); Citeseer $p = 0.053$. Linear and Kendall schedules match the controller on accuracy but do not enforce the budget at the best checkpoint; GradNorm enforces it already at the best checkpoint and pays 1.1–1.8 points. The controller enforces the budget late in training without accuracy loss; late-phase similarity was not logged for the three baselines (Table 2).

its weight saturates at the renormalization bound ($\lambda_{\text{final}} \approx 1.95$ on every dataset; on Pubmed it saturates without moving similarity at all). The linear and Kendall schedules do not enforce the budget at the best checkpoint ($d_{\text{best}} \geq 0.849$ everywhere) and stay harmless. On Cora, the controller enforces the budget late, after the best checkpoint, while matching baseline accuracy. Late-phase similarity was logged only for the controller; the baselines' timing is therefore judged by best-checkpoint similarity alone, a protocol limitation we state in Sec. 5.

Baseline adaptations. GradNorm is designed for multi-task balance; we use its two-term form with the CE loss and $\mathcal{L}_{\text{div}}$ as the two "tasks": each epoch, the gradient norms of both terms with respect to the shared weights are measured, and the divergence weight is nudged toward equalizing the normalized norms (asymmetry $\alpha = 0$, step $\eta = 0.16$); both weights are then renormalized to sum to 2. The saturation at $\lambda \approx 1.95$ follows from the combination of a 0.05 floor on the CE weight with the sum-to-2 renormalization, which bounds the divergence weight at $2 - 0.05 = 1.95$. The Kendall–Gal baseline treats $\mathcal{L}_{\text{div}}$ as a second "task" with a learnable log-variance: $L = e^{-s_{\text{CE}}} L_{\text{CE}} + e^{-s_{\text{div}}} \mathcal{L}_{\text{div}} + (s_{\text{CE}} + s_{\text{div}})/2$, and we report the effective weight $\lambda = e^{-s_{\text{div}}}$.

Table 2. Controller vs. adaptive baselines (one-sided paired tests, 10 seeds). Accuracy is mean ± std per cell; λ_{final} and d_{best} are the mean final controller weight and the inter-head similarity at the best checkpoint. GradNorm is the only baseline whose best-checkpoint similarity reaches the budget, at a significant accuracy cost; linear and Kendall schedules stay accuracy-neutral and do not enforce the budget at the best checkpoint. The controller enforces the budget late in training while matching baseline accuracy (late-phase similarity was not logged for the three baselines).

Dataset	Schedule	Test acc. (%)	λ_{final}	d_{best}	Δ vs. ctrl. (pp)	p (paired)
Cora	Controller	80.79 ± 0.92	0.84	0.934	—	—
	Linear	80.83 ± 1.05	0.87	0.924	−0.04	0.4531
	Kendall	80.83 ± 1.21	1.37	0.849	−0.04	0.5781
	Gradnorm	79.04 ± 1.57	1.95	0.693	+1.75	0.0098
Citeseer	Controller	68.46 ± 1.35	0.27	0.988	—	—
	Linear	68.94 ± 1.01	0.30	0.976	−0.48	0.7764
	Kendall	68.66 ± 0.90	1.10	0.969	−0.20	0.6055
	Gradnorm	67.36 ± 1.35	1.95	0.688	+1.10	0.0527
Pubmed	Controller	77.44 ± 0.67	1.94	0.996	—	—
	Linear	77.23 ± 0.79	1.67	0.995	+0.21	0.3477
	Kendall	77.32 ± 0.76	1.26	0.996	+0.12	0.3389
	Gradnorm	76.07 ± 0.67	1.95	1.000	+1.37	0.0049

4.3. *Does the diversity criterion beat standard pruning criteria?*

Claim tested: the $\mathcal{L}_{\text{div}}$ similarity signal selects better heads than gradient/random/magnitude.

Under adaptive teachers at prune ratio 0.75 (10 paired seeds, fine-tuning 100 epochs, no KD), all contrasts are null: vs. random +0.05 pp (one-sided Wilcoxon $p = 0.28$), vs. gradient −0.13 pp ($p = 0.63$), vs. magnitude +0.01 pp ($p = 0.54$). At ratio 0.5 and under B0 (unregularized) teachers the pattern repeats: of the nine remaining cells (Adaptive ratio 0.5 and B0 ratios 0.5 and 0.75, three criteria each), eight have one-sided Wilcoxon $p = 0.50$–1.00, with one exception: under the B0 teacher at ratio 0.5, diversity vs. gradient shows +0.62 pp (Wilcoxon $p = 0.094$, paired-t $p = 0.048$, $d = 0.97$). This does not survive multiplicity control (Holm–Bonferroni across the 12 contrasts: the smallest p-value, 0.048, is above the corrected threshold $0.05/12 \approx 0.0042$, so no contrast survives) and is directionally inconsistent with the main comparison. The ratio-0.5 and B0-teacher cells come from the pre-submission batch and use $n = 5$ seeds per arm (5 pairs); the adaptive-teacher ratio-0.75 contrasts use the 10-seed revision batch. We report the full matrix in Table 3 and conclude: at GAT scale with 100-epoch fine-tuning, the similarity criterion performs on par with strong baselines, and we find no evidence that it is better.

Mechanism. The criterion is computed at the best checkpoint, where head similarity is high for every configuration (0.93–0.99). At the point where pruning happens, the $\mathcal{L}_{\text{div}}$ signal does not differentiate heads. This is consistent with:[14] fine-tuning washes out criterion choice.

Table 3. Paired comparison of pruning criteria at GAT scale (fine-tuning 100 epochs, no KD; identical seeds within each contrast). Δ is the accuracy difference (diversity criterion minus baseline criterion) in percentage points; Cohen's d is reported for each contrast. Wilcoxon p-values are one-sided (diversity > baseline). No contrast survives multiplicity control (Holm–Bonferroni across the 12 tests); the single nominally significant pair (B0 teacher, ratio 0.5, vs. gradient) rests on 5 pairs (these pre-submission cells use n=5 seeds per arm; the adaptive-teacher ratio-0.75 cells use the 10-seed revision batch) and is directionally inconsistent with the main comparison. Tied pairs in the Wilcoxon test fall back to the normal approximation; paired-t p-values are reported alongside.

Teacher	Ratio	Contrast	Δ (pp)	Wilcoxon p	Paired-t p	d
Adaptive	0.5	div vs. random	-1.20	1.0000	0.9828	-1.41
Adaptive	0.5	div vs. gradient	-0.22	0.7812	0.6893	-0.24
Adaptive	0.5	div vs. magnitude	-0.26	0.8125	0.7677	-0.36
Adaptive	0.75	div vs. random	$+0.05$	0.2783	0.4466	+0.04
Adaptive	0.75	div vs. gradient	-0.13	0.6289	0.7593	-0.23
Adaptive	0.75	div vs. magnitude	$+0.01$	0.5391	0.4807	+0.02
B0	0.5	div vs. random	-0.34	0.7812	0.8067	-0.43
B0	0.5	div vs. gradient	$+0.62$	0.0938	0.0485	+0.97
B0	0.5	div vs. magnitude	-0.20	0.5938	0.6924	-0.24
B0	0.75	div vs. random	-0.48	0.9062	0.8254	-0.47
B0	0.75	div vs. gradient	$+0.04$	0.5000	0.4778	+0.03
B0	0.75	div vs. magnitude	$+0.12$	0.6875	0.4110	+0.11

Revival test (Table 4). The negative result could be an artifact of this timing: the controller moves similarity late in training (Sec. 4.2), so a criterion computed at a late checkpoint might recover the signal. We therefore pruned from three checkpoint sources (best, lowest-similarity, final) with all four criteria, 10 seeds per cell, on all datasets (prune ratio 0.75, fine-tuning 100 epochs, no KD). The lowest-similarity checkpoint coincides with the final one in 119 of 120 combinations, since similarity decreases monotonically under the controller, so the late source carries the differentiated signal ($d_{\text{src}} = 0.71$–0.85 on Cora and Citeseer). All nine late-source contrasts remain null (one-sided $p \geq 0.078$; the nominally largest effect favors random, -0.87 pp on Pubmed), and pruning from the best checkpoint is never significantly worse than pruning from the late one; on Citeseer the best checkpoint is significantly better ($+1.90$ pp, $p = 0.001$). The negative criterion result is therefore not a timing artifact: even where the $\mathcal{L}_{\text{div}}$ signal differentiates heads, 100 epochs of fine-tuning erase criterion choice. On Pubmed the probe is weak: the teacher's late similarity stays at 0.99, because the run is response-limited there (Sec. 4.2 shows even a $4\times$ gain saturates the ceiling without moving similarity), so the decisive evidence is Cora and Citeseer.

4.4. *Does KD recover pruning loss, and how does prune+KD compare to pure KD?*

Claim tested: KD restores pruned-model accuracy; against pure KD at the same head budget, no significant difference is detectable.

Table 4. Revival test: pruning criteria computed from late-training checkpoints. Top: test accuracy (mean ± std) by checkpoint source and criterion. d_{src} is the teacher's head similarity at the checkpoint source. The lowest-similarity checkpoint coincides with the final one in 119/120 (dataset, seed, criterion) combinations (similarity decreases monotonically), so d_{src} for the late source is the teacher's final similarity. Bottom: one-sided paired contrasts (diversity vs. baseline criterion) at the late source, the timing where the signal is actually differentiated; none is significant, and the nominally largest effect favors random. The negative criterion result therefore survives the revival test.

Dataset	Source	d_{src}	Diversity	Magnitude	Gradient	Random
Cora	Best	0.934	80.32 ± 0.78	80.31 ± 0.59	80.45 ± 0.66	80.27 ± 0.75
	Late (min div.)	0.706	80.36 ± 1.49	79.95 ± 0.86	79.88 ± 1.01	79.74 ± 1.05
	Late (last)	0.706	80.36 ± 1.49	79.95 ± 0.86	79.88 ± 1.01	79.74 ± 1.05
Citeseer	Best	0.972	67.95 ± 1.17	67.94 ± 1.13	67.77 ± 1.36	68.35 ± 1.05
	Late (min div.)	0.847	66.05 ± 1.35	66.21 ± 1.80	66.65 ± 1.67	66.87 ± 1.19
	Late (last)	0.847	66.05 ± 1.35	66.18 ± 1.86	66.65 ± 1.67	66.87 ± 1.19
Pubmed	Best	0.995	76.35 ± 0.93	76.33 ± 1.02	77.02 ± 0.72	76.77 ± 1.51
	Late (min div.)	0.988	76.17 ± 1.25	75.98 ± 1.03	76.51 ± 0.76	77.04 ± 1.31
	Late (last)	0.988	76.17 ± 1.25	75.98 ± 1.03	76.51 ± 0.76	77.04 ± 1.31

Paired contrasts at the late source (diversity minus baseline, 10 pairs)

Dataset	Contrast	Δ (pp)	Wilcoxon p	Paired-t p	d
Cora	vs. random	$+0.62$	0.1162	0.1153	0.41
Cora	vs. magnitude	$+0.41$	0.0781	0.0779	0.49
Cora	vs. gradient	$+0.48$	0.1436	0.1223	0.39
Citeseer	vs. random	-0.82	0.9502	0.9475	-0.57
Citeseer	vs. magnitude	-0.16	0.6289	0.6122	-0.09
Citeseer	vs. gradient	-0.60	0.8691	0.8296	-0.32
Pubmed	vs. random	-0.87	0.9756	0.9589	-0.62
Pubmed	vs. magnitude	$+0.19$	0.2656	0.1509	0.35
Pubmed	vs. gradient	-0.34	0.8203	0.8391	-0.33

At both budgets, KD recovers most of the pruning loss: B8 (prune+KD) vs. B6 (prune+FT) gains +2.11 pp at 4/8 heads and +2.38 pp at 2/8 heads (one-sided Wilcoxon $p = 0.001$, $d \approx 2.1$, 10 pairs). Against pure KD (B5: small student trained from scratch with KD, matched 100-epoch budget), B8 is numerically slightly lower but not significantly so: -0.48 pp (95% CI $[-1.12, +0.16]$, two-sided Wilcoxon $p = 0.19$) at ratio 0.5 and -0.84 pp (CI $[-1.79, +0.11]$, $p = 0.13$) at ratio 0.75. With 10 pairs and $|d| \leq 0.84$ pp the test is underpowered, so we claim only that no significant difference is detectable, not equivalence. The two paths also cost the same wall time on a T4 (7.9 ± 1.1 s vs. 8.8 ± 0.1 s per run at ratio 0.5; 8.4 ± 0.1 s vs. 8.3 ± 0.1 s at ratio 0.75; B5 vs. B8, both including dense teacher training): the pruning path reuses the trained dense weights instead of re-initializing a small student, but offers no wall-time saving at this scale. Pure KD alone already captures the recovery; the structured (pruned) student reaches comparable accuracy through the pruning path. A secondary observation: pure-KD students exceed the dense teacher (82.5–82.9 vs. 81.15 on Cora), consistent with the student-over-teacher phenomenon reported in.[21]

What does the pruning path cost? The missing reference is a small model trained from scratch without KD (B4, same budgets, 100 epochs). On Cora, B4 reaches 80.73±1.36 pp at 4/8 heads and 80.60±1.05 pp at 2/8 heads, so prune+fine-tuning (B6) sits 0.8–0.9 pp *below* the from-scratch small model: the fine-tuning gap (2.1–2.4 points below B8) is not just a property of the small architecture. Pruning followed by plain fine-tuning is a small net loss on Cora relative to training the matched small architecture from scratch, and KD recovers it. The same four arms on Citeseer and Pubmed (Appendix C) do not show this loss: B6 is at or above B4 in three of the four extension cells (Citeseer +0.65 pp at ratio 0.75; Pubmed +0.24 pp at 0.5 and +1.36 pp at 0.75; Citeseer at 0.5: −0.25 pp). There, pruning inherits the dense teacher's initialization before fine-tuning, which appears to offset the removed heads, so the Cora finding should be read as dataset-specific rather than as a general property of pruning. Pruning+KD (B8) matches pure KD, which is itself the ceiling at this budget.

Table 5. Knowledge distillation at fixed head budgets (Cora; mean ± std over 10 seeds; teacher = adaptive dense model). B5: small student trained from scratch with KD. B6: diversity-based pruning + fine-tuning (100 epochs). B8: pruning + KD (100 epochs). Contrasts are paired Wilcoxon signed-rank tests over identical seeds; Cohen's d in parentheses. B8 vs. B5 is statistically tied at both budgets.

Ratio	B5 (pure KD)	B6 (prune+FT)	B8 (prune+KD)	B8 − B6	B8 − B5
0.5	82.48 ± 1.08	79.89 ± 0.64	82.00 ± 0.69	+2.11 (p=0.001, d=2.06)	−0.48 (p=0.92, d= − 0.53)
0.75	82.88 ± 1.16	79.66 ± 1.56	82.04 ± 1.46	+2.38 (p=0.001, d=2.07)	−0.84 (p=0.95, d= − 0.63)

5. Discussion and Limitations

What the controller does and does not do. The adaptive controller enforces the similarity budget late in training while preserving accuracy; the first feedback-controlled diversity regularizer we are aware of. The budget is met on Cora, partially met on Citeseer, and response-limited on Pubmed (Sec. 4.2); this dataset dependence is the main caveat on the claim's generality. The sensitivity grid in Appendix B shows accuracy is stable across (η, d^*) on Cora; it does not transfer that promise to Pubmed, where a 4× gain saturates the ceiling without moving similarity (Sec. 4.2). The controller does not reduce similarity at the best checkpoint (d remains ∼0.93 on Cora), which is exactly why it is harmless: the divergence happens where the task loss is already flat. This restates the trade-off in fixed-λ regularizers: their failure is not that diversity is harmful, but that a constant weight cannot target the late-training phase.

Uses of the controlled quantity. Within the scope of this paper, the honest answer is that we have not yet converted the controlled late-training similarity into a measurable downstream gain: at GAT scale the pruning criterion built on it does not beat standard criteria, and prune+KD only matches pure KD. We therefore

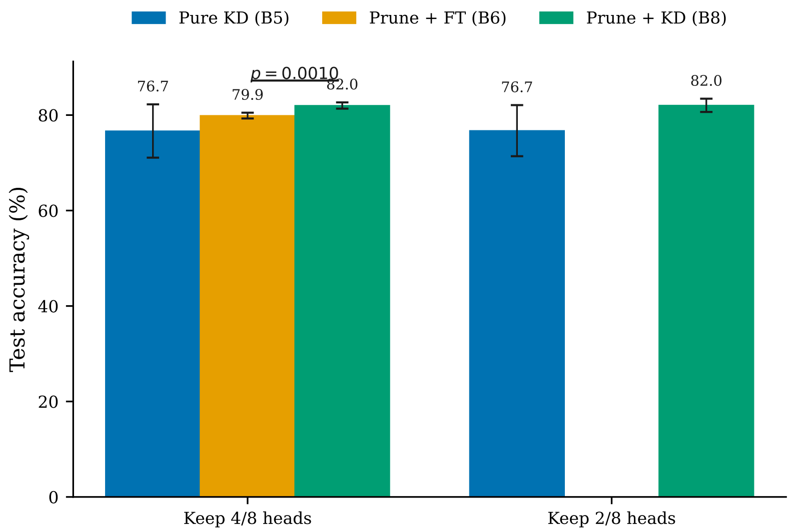

Fig. 4. Test accuracy at fixed head budgets: training a smaller student from scratch with KD (B5), diversity-based pruning followed by fine-tuning (B6), and pruning followed by KD (B8), at prune ratios 0.5 (4/8 heads) and 0.75 (2/8 heads). Bars are means over 10 seeds; error bars are ± 1 standard deviation. Brackets: one-sided paired Wilcoxon signed-rank test, B8 vs. B6, $p = 0.001$ at both ratios. B8 vs. B5 shows no significant difference (two-sided Wilcoxon $p = 0.19$ at ratio 0.5, $p = 0.13$ at ratio 0.75; 95% CIs on the difference in the text); KD, not the pruning path, accounts for the recovery.

state the settings where the control is, based on our evidence, applicable, and one where it is not. The controller's output, a final checkpoint with low inter-head similarity and an early checkpoint with high similarity, is the same kind of signal that inference-time head fusion and clustering methods (CHAI, DHA, ProxyAttn; Sec. 2) exploit; those methods operate on the *final* checkpoint, where fixed weights either never moved similarity ($\lambda \leq 1$) or already paid for it ($\lambda \geq 5$), whereas the controller produces the low-similarity checkpoint at no accuracy cost. We view head-fusion loss on controller checkpoints as the natural downstream payoff of the controlled quantity, and plan to test it in future work. A second applicable setting is any pipeline that needs a hard similarity constraint at a late checkpoint (for example, before an inference-time merge step), which the integral law provides without hand-tuning per dataset. The setting where it is *not* applicable is improving a single model's test accuracy: the controller is no-harm by design and by evidence, not an accuracy booster.

The negative criterion result. We expected the shared-metric design (regularize with $\mathcal{L}_{\text{div}}$, prune with $\mathcal{L}_{\text{div}}$) to give the similarity criterion an advantage. It does not, at GAT scale. The dual-use idea is appealing and appears in related work,[2] but our strict paired tests show fine-tuning dominates criterion choice; we report the negative evidence accordingly. The revival test (Sec. 4.3) extends this to late-training checkpoints, where the signal is actually differentiated: the null persists, so the negative result is not a best-checkpoint artifact.

Limitations. (1) Single metric family: all results use attention-coefficient cosine similarity; other diversity measures are untested. (2) GAT-only evidence: pruning and distillation results do not cover Transformer-scale architectures; transfer is untested. (3) Best-checkpoint similarity remains high (Sec. 4.2); divergence is a training-time dynamic, not a checkpoint property. Late-phase similarity was logged only for the controller; the timing of the adaptive baselines is judged by best-checkpoint similarity alone. (4) The pre-submission Cora adaptive-dense 10-seed batch ran on Colab T4 without archived per-run Hydra overrides; the protocol and seeds are documented, and the results match same-protocol local runs, but per-run config archives for that batch are unavailable. Some pre-submission cells (the Cora fixed-λ sweep and the ratio-0.5/B0 criterion cells) used $n = 5$ seeds; the second revision re-ran the sweep at $n = 10$ with recorded similarities and supersedes the earlier numbers (Table 1). The revision experiments (adaptive baselines, controller checkpoints, and the revival matrix) were executed from a versioned notebook with per-combination progress logging, and their merged results and teacher checkpoints are archived. (5) B8-vs-B5 power is limited (10 seeds, $|d| \leq 0.84$ pp, 95% CIs on the difference including zero at both ratios); we claim only that no significant difference is detectable, not equivalence. (6) Code, protocols, seeds, and merged results are available at `https://github.com/HuKejie/div-prune`.

6. Conclusion

Fixed-weight diversity regularization of multi-head attention has no operating point that is both effective and harmless: the weakest effective weight ($\lambda = 5$) already enforces diversity at the best checkpoint and pays about 0.7 points for it. A feedback controller with a similarity budget achieves what no fixed weight can: the budget is enforced *late* in training, after the best checkpoint, while accuracy stays within ± 0.4 points of baseline. The late phase gets there on Cora (similarity 0.71), substantially on Citeseer (0.85), and not on Pubmed, where the similarity response is too slow; we document this dataset dependence rather than hide it (Sec. 4.2, Appendix B). Against adaptive baselines, the controller enforces the budget late while preserving accuracy; GradNorm enforces it at the best checkpoint and pays 1.1–1.8 points for it, and the linear and Kendall schedules match the controller's accuracy without enforcing the budget at the best checkpoint (Sec. 4.2). The same similarity metric, used as a pruning criterion, performs on par with standard criteria, a strict negative result that survives a revival test at late-training checkpoints (Sec. 4.3); distillation restores pruning loss to the level of students trained from scratch with KD at the same budget (no significant difference, Sec. 4.4). On Pubmed at ratio 0.75, prune+KD (77.95) exceeds the from-scratch small model without KD (74.99) by about 3 points; we report this as a data point, not a claimed advantage, since it may combine the teacher initialization with the KD effect. The practical lesson: for small multi-head models, control the regularizer, not the hyperparameter; and pair any pruning criterion with KD, since KD dominates criterion choice at this scale.

References

1. S. Agarwal, B. Acun, B. Hosmer, M. Elhoushi, Y. Lee, S. Venkataraman, D. Papailiopoulos and C.-J. Wu, Chai: Clustered head attention for efficient llm inference (2024).
2. N. Ampazis and F. Sakketou, Diversifying multi-head attention in the transformer model, *Machine Learning and Knowledge Extraction* **6** (November 2024) p. 2618–2638.
3. K. Audhkhasi, A. Rosenberg and B. Ramabhadran, Analysis of self-attention head diversity for conformer-based automatic speech recognition, in *Proc. Interspeech 2022* (ISCA, 2022).
4. D. Chen, N. Liu, Y. Zhu, Z. Che, R. Ma, F. Zhang, X. Mou, Y. Chang and J. Tang, Epsd: Early pruning with self-distillation for efficient model compression (2024).
5. T. Chen, Z. Zhang, Y. Cheng, A. Awadallah and Z. Wang, The principle of diversity: Training stronger vision transformers calls for reducing all levels of redundancy (2022).
6. Y. Chen, L. Zhang, J. Shang, Z. Zhang, T. Liu, S. Wang and Y. Sun, Dha: Learning decoupled-head attention from transformer checkpoints via adaptive heads fusion (2024).
7. Z. Chen, V. Badrinarayanan, C.-Y. Lee and A. Rabinovich, Gradnorm: Gradient normalization for adaptive loss balancing in deep multitask networks (2018).
8. E. Harvey, M. Petrov and M. C. Hughes, Learning the regularization strength for deep fine-tuning via a data-emphasized variational objective (2025).
9. Y. He, S. Zhou, G. Zhang, H. Yun, Y. Xu, B. Zeng, T. Chilimbi and H. Zhao, Robust multi-task learning with excess risks (2024).
10. G. Hinton, O. Vinyals and J. Dean, Distilling the knowledge in a neural network (2015).
11. N. K. Jha and B. Reagen, Entropy-guided attention for private llms (2025).
12. X. Jiao, Y. Yin, L. Shang, X. Jiang, X. Chen, L. Li, F. Wang and Q. Liu, Tinybert: Distilling bert for natural language understanding (2020).
13. A. Kendall, Y. Gal and R. Cipolla, Multi-task learning using uncertainty to weigh losses for scene geometry and semantics (2018).
14. P. Michel, O. Levy and G. Neubig, Are sixteen heads really better than one? (2019).
15. A. M. Nicolicioiu, A. L. Nicolicioiu, B. Alexe and D. Teney, Learning diverse features in vision transformers for improved generalization (2023).
16. J. Park and A. No, Prune your model before distill it (2022).
17. P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò and Y. Bengio, Graph attention networks (2018).
18. E. Voita, D. Talbot, F. Moiseev, R. Sennrich and I. Titov, Analyzing multi-head self-attention: Specialized heads do the heavy lifting, the rest can be pruned (2019).
19. W. Wang, F. Wei, L. Dong, H. Bao, N. Yang and M. Zhou, Minilm: Deep self-attention distillation for task-agnostic compression of pre-trained transformers (2020).
20. Y. Wang, H. He, S. Bao, H. Wu, H. Wang, Q. Zhu and W. Che, Proxyattn: Guided sparse attention via representative heads (2026).
21. L. Wu, H. Lin, Y. Huang and S. Z. Li, Quantifying the knowledge in gnns for reliable distillation into mlps (2023).
22. Z. Yang, W. W. Cohen and R. Salakhutdinov, Revisiting semi-supervised learning with graph embeddings (2016).
23. H. Yun, T. Kang and K. Jung, Analyzing and controlling inter-head diversity in

multi-head attention, *Applied Sciences* **11** (February 2021) p. 1548.
24. S. Zhang, Y. Liu, Y. Sun and N. Shah, Graph-less neural networks: Teaching old mlps new tricks via distillation (2022).
25. Z. Zhang, Y. Lu, T. Wang, X. Wei and Z. Wei, Ddk: Dynamic structure pruning based on differentiable search and recursive knowledge distillation for bert, *Neural Networks* **173** (May 2024) p. 106164.

Appendix A. 95% confidence intervals

Table 6. Main-comparison cells with 95% confidence intervals (10 seeds).

Dataset	Baseline	Fixed λ=20	Adaptive
Cora	$[80.42, 81.88]$	$[78.82, 80.60]$	$[80.13, 81.45]$
Citeseer	$[67.15, 69.39]$	$[65.12, 67.98]$	$[67.49, 69.43]$
Pubmed	$[77.13, 77.91]$	$[75.81, 76.79]$	$[76.96, 77.92]$

Appendix B. Controller sensitivity to η and d^*

Appendix C. H3 extension to Citeseer and Pubmed

Table 9 repeats the four KD-pipeline arms on Citeseer and Pubmed. The KD-recovery pattern is consistent across datasets: B8 (prune+KD) beats B6 (prune+FT) at every cell, significantly so in three of the four extension cells under one-sided Wilcoxon tests with Holm–Bonferroni correction across the four cells (Citeseer +1.25 pp, $p = 0.006$, at ratio 0.75; Pubmed +0.77 pp, $p = 0.005$, at ratio 0.5; Pubmed +1.60 pp, $p = 0.001$, at ratio 0.75; the Citeseer ratio-0.5 cell gives +0.63 pp, $p = 0.161$). Against pure KD (B5), B8 shows no significant difference in three of four cells (two-sided tests, 95% CIs on the difference including zero: Citeseer −0.41 pp $[-1.43, +0.61]$, $p = 0.49$, at 0.5 and −0.79 pp $[-2.20, +0.62]$, $p = 0.28$, at 0.75; Pubmed −0.29 pp $[-0.80, +0.22]$, $p = 0.19$, at 0.5). On Pubmed at ratio 0.75 B8 is slightly above B5 (+0.62 pp, 95% CI $[+0.28, +0.96]$, $p = 0.006$, which survives the Holm correction); we do not claim this generalizes, and note it is consistent with the student-over-teacher observation already reported for Cora (Sec. 4.4).

Table 7. 95% confidence intervals for the revision protocol (10 seeds per cell). Top: adaptive-baseline comparison (R1); controller rows are the submitted dense adaptive block. Bottom: revival test (R2), test accuracy by checkpoint source and criterion; prune ratio 0.75, fine-tuning 100 epochs.

R1: adaptive baselines				
Dataset	Schedule	Test acc. 95% CI	λ_{final}	d_{best}
Cora	Controller	80.79 [80.13, 81.45]	0.84	0.934
	Linear	80.83 [80.08, 81.58]	0.87	0.924
	Kendall	80.83 [79.97, 81.69]	1.37	0.849
	Gradnorm	79.04 [77.91, 80.17]	1.95	0.693
Citeseer	Controller	68.46 [67.49, 69.43]	0.27	0.988
	Linear	68.94 [68.22, 69.66]	0.30	0.976
	Kendall	68.66 [68.01, 69.31]	1.10	0.969
	Gradnorm	67.36 [66.40, 68.32]	1.95	0.688
Pubmed	Controller	77.44 [76.96, 77.92]	1.94	0.996
	Linear	77.23 [76.67, 77.79]	1.67	0.995
	Kendall	77.32 [76.78, 77.86]	1.26	0.996
	Gradnorm	76.07 [75.59, 76.55]	1.95	1.000

R2: revival test (test accuracy 95% CIs)					
Dataset	Source	Diversity	Magnitude	Gradient	Random
Cora	Best	80.32 [79.76, 80.88]	80.31 [79.89, 80.73]	80.45 [79.98, 80.92]	80.27 [79.73, 80.81]
	Late (min div.)	80.36 [79.30, 81.42]	79.95 [79.34, 80.56]	79.88 [79.16, 80.60]	79.74 [78.99, 80.49]
	Late (last)	80.36 [79.30, 81.42]	79.95 [79.34, 80.56]	79.88 [79.16, 80.60]	79.74 [78.99, 80.49]
Citeseer	Best	67.95 [67.11, 68.79]	67.94 [67.13, 68.75]	67.77 [66.80, 68.74]	68.35 [67.60, 69.10]
	Late (min div.)	66.05 [65.08, 67.02]	66.21 [64.93, 67.49]	66.65 [65.46, 67.84]	66.87 [66.02, 67.72]
	Late (last)	66.05 [65.08, 67.02]	66.18 [64.85, 67.51]	66.65 [65.46, 67.84]	66.87 [66.02, 67.72]
Pubmed	Best	76.35 [75.69, 77.01]	76.33 [75.60, 77.06]	77.02 [76.51, 77.53]	76.77 [75.69, 77.85]
	Late (min div.)	76.17 [75.27, 77.07]	75.98 [75.25, 76.71]	76.51 [75.97, 77.05]	77.04 [76.11, 77.97]
	Late (last)	76.17 [75.27, 77.07]	75.98 [75.25, 76.71]	76.51 [75.97, 77.05]	77.04 [76.11, 77.97]

Table 8. Controller sensitivity to (η, d^*) on Cora (10 seeds per cell; mean $\pm$ std). d_{best} is the mean inter-head similarity at the best checkpoint, "Ep." the mean last epoch run under early stopping, and λ_{final} the mean final weight. The controller acts *late* in training, so d_{best} responds only weakly to the grid; the budget is enforced after the best checkpoint (the main arm's late-training similarity is 0.706 on Cora, Table 4). The $(0.05, 0.7)$ cell is the submitted main arm (original batch); the other eight cells are the second-revision grid batch (identical protocol).

η	d^*	Test acc. (%)	d_{best}	Ep.	λ_{final}
0.01	0.5	81.04 ± 1.62	0.966	150	0.29
0.01	0.7	80.98 ± 1.45	0.974	151	0.17
0.01	0.9	81.16 ± 1.43	0.978	150	0.05
0.05	0.5	80.94 ± 1.50	0.906	152	1.24
0.05	0.7	80.79 ± 0.92	0.934	160	0.84
0.05	0.9	81.04 ± 1.62	0.971	150	0.23
0.2	0.5	80.53 ± 1.45	0.853	149	2.78
0.2	0.7	80.72 ± 1.53	0.884	148	2.19
0.2	0.9	81.06 ± 1.47	0.949	146	0.53

Table 9. The four KD-pipeline arms at matched head budgets on Citeseer and Pubmed (10 seeds; mean $\pm$ std). B4: from-scratch small model, no KD. B5: pure KD from scratch. B6: prune + fine-tuning. B8: prune + KD. Cora cells repeat the submitted H3 comparison (Table 5); paired statistics for the Citeseer/Pubmed cells are reported in the Appendix C text.

Dataset	Ratio	B4 (scratch)	B5 (pure KD)	B6 (prune+FT)	B8 (prune+KD)
Cora	0.5	80.73 ± 1.36	82.48 ± 1.08	79.89 ± 0.64	82.00 ± 0.69
Cora	0.75	80.60 ± 1.05	82.88 ± 1.16	79.66 ± 1.56	82.04 ± 1.46
Citeseer	0.5	68.47 ± 1.56	69.26 ± 1.14	68.22 ± 0.75	68.85 ± 1.37
Citeseer	0.75	67.30 ± 1.53	69.99 ± 1.07	67.95 ± 1.17	69.20 ± 1.24
Pubmed	0.5	76.95 ± 0.74	78.25 ± 0.52	77.19 ± 0.86	77.96 ± 0.72
Pubmed	0.75	74.99 ± 0.92	77.33 ± 0.31	76.35 ± 0.93	77.95 ± 0.38